

Firezer Demese

Senior SQL Server Database Administrator | AI-Enhanced Operations | Performance Engineering | HA/DR | Automation
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PROFESSIONAL SUMMARY

Senior SQL Server Database Administrator with 7+ years delivering enterprise-scale database reliability, performance, and intelligent automation. Currently at **TransCore** managing 150+ production SQL Server instances processing 2.5 million transactions daily, and previously at **Bank of America** managing servers in a regulated financial services environment. And before that at **Shell managing Oil and Gas operations**. Combining expertise in SQL Server Always On AG, performance tuning, and PowerShell automation with **AI-enhanced database operations** - building intelligent automation frameworks, **LLM-powered SQL analysis tools**, and predictive monitoring systems that catch and resolve incidents before users feel them. Actively developing AI-driven DBA tooling using Python, FastAPI, and the Claude API to deliver self-healing database operations at scale.

TECHNICAL SKILLS

SQL Server	(2012-2022) - enterprise administration, performance, security, and reliability at scale
High Availability and DR	Always On Availability Groups, Windows Server Failover Clustering, Log Shipping, Replication, DR planning and testing
Performance Optimization	Execution plan analysis, DMVs, Query Store, wait statistics, index optimization, parameter sniffing resolution, blocking and deadlock resolution
AI-Enhanced Ops	LLM integration via Claude API and OpenAI GPT for SQL analysis and NL-to-SQL; Python anomaly detection; intelligent automation and self-healing database operations
Automation	PowerShell - health checks, AG monitoring, backup compliance, disk alerts, job failure detection; T-SQL stored procedures and SQL Agent jobs; Python scripting
Backup and Recovery	Full, differential, and log backup strategy; Azure Blob Storage ; Wasabi Immutable air-gap repositories; DR rehearsals; RTO/RPO compliance
BI and Data	SSIS ETL pipeline troubleshooting; SSRS operational and financial reporting; SSAS, GoldenGate
Monitoring	Redgate SQL Monitor, SQL Server Profiler, Extended Events, native DMVs, custom PowerShell monitoring frameworks, proactive capacity planning
Cloud	Azure SQL Database, Azure SQL Managed Instance, Azure Blob Storage, AWS RDS, hybrid on-premises and cloud environments
Security	SOX and HIPAA compliance - TDE encryption; RBAC and least-privilege enforcement; SQL Server Audit logging; change management

AI-POWERED DBA PROJECTS (IN DEV)

QueryLens - LLM-Powered SQL Performance Analyzer

- Full-stack web application using Claude API and OpenAI GPT to convert natural language questions to optimized SQL and analyze existing queries for performance anti-patterns.
- Detects 9 anti-pattern categories with quality scoring and actionable fix recommendations.
- Executes generated SQL against live PostgreSQL and returns structured results - **bridging traditional DBA expertise with AI-powered data tooling**.
- Stack:** Python, FastAPI, PostgreSQL, Claude API / OpenAI API, HTML/CSS/JavaScript, Docker, Railway deployment.

SQLGuardian - Intelligent SQL Server Health Monitoring and Auto-Remediation Engine

- Python-based monitoring engine collecting SQL Server health metrics via DMVs across production instances - AG health, backup compliance, disk space, blocking chains, job failures.
- Statistical anomaly detection using baseline comparison to identify deviations before users feel impact; auto-remediation for blocking chains, emergency log backups, and index rebuild scheduling.
- Exposes REST API via FastAPI for external tool integration; LLM integration (Claude API) for automated plain-English incident explanation and remediation recommendations.
- **Stack:** Python, FastAPI, PowerShell, SQL Server DMVs, asyncpg, Docker, GitHub Actions CI/CD.

WORK EXPERIENCE

TransCore

Jan 2023 - Present

Senior SQL Server Database Administrator | Houston, TX (Remote)

Responsible for the reliability, performance, and security of 150+ production SQL Server instances supporting nationwide applications - databases up to 6TB, approximately 2.5 million transactions per day.

- Administer, maintain & install enterprise SQL Server 2016-2022 production environments - availability, performance, capacity planning, and security compliance across 150+ instances.
- Design and manage Always On Availability Groups with synchronous and asynchronous secondary replicas. Monitor replica health via DMVs and custom PowerShell alerts. Maintain immutable air-gap backup repositories (Wasabi) for ransomware protection alongside Azure Blob Storage backups.
- Perform advanced performance tuning using execution plan analysis, Query Store, wait statistics, and index optimization - achieved 60% average reduction in query runtime.
- Build and maintain PowerShell automation frameworks for health checks, AG monitoring, backup compliance, disk space management, and blocking chain alerting - saving the team 12+ hours per week.
- Install, configure, and maintain SSIS ETL pipelines and SSRS reporting suites. Troubleshoot package failures and report performance issues.
- Implement multi-tier backup strategy: local disk, Azure Blob Storage, and immutable air-gap cloud storage. Monitor compliance every 15 minutes via automated PowerShell validation.
- Maintain database security controls including TDE, RBAC, least-privilege access, and auditing. Zero SOX and HIPAA audit findings.
- Mentored junior DBAs on operational standards, runbook documentation, and incident response processes.

Bank of America

Jun 2020 - Dec 2022

SQL Database Administrator | Houston, TX

Database administrator supporting high-throughput banking platforms in a SOX-regulated environment, responsible for platform reliability, performance optimization, security compliance, and automation.

- Built Python and PowerShell automation frameworks for health monitoring, backup management, alerting, and operational workflows - reduced manual triage time by 30%.
- Managed SQL Server production databases supporting financial applications under strict SLA and SOX compliance requirements - zero audit findings across all quarterly reviews.
- Diagnosed and resolved a critical blocking chain affecting 200+ concurrent users within the SLA window, then implemented proactive monitoring.
- Maintained Always On AG and Log Shipping for HA/DR. Led quarterly DR failover exercises to validate recovery time objectives.
- Enforced SOX-aligned database security: TDE, RBAC, least-privilege access, SQL Server Audit logging to write-once storage, and formal change management with peer review and rollback plans.
- Collaborated with application development, Infrastructure and data teams to optimize query performance for trading, analytics, and operational workloads.

Shell

Feb 2019 - May 2020

Database Administrator | Houston, TX

Database administrator supporting SQL Server environments for oil and gas operations across Windows and Linux infrastructure.

- Administered SQL Server production databases across dev, QA, and production environments. Managed patching, upgrades, and capacity planning.
- Executed a SQL Server 2016 to 2019 migration using Data Migration Assistant for compatibility assessment - side-by-side approach with zero production downtime.
- Developed PowerShell and Bash automation scripts for deployment workflows, health monitoring, and alerting.
- Implemented Always On AG and Log Shipping HA/DR configurations. Executed DR drills and maintained failover readiness.

EDUCATION AND CERTIFICATIONS

Microsoft Certified: Azure Database Administrator Associate - DP-300 (Pursuing)

Python for Data Science - Great Learning

Bachelor of Science, Computer Science - Addis Ababa University